

SAI KRISHNA REDDY

Senior Software Engineer · AI Systems · Distributed Data Infrastructure

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SUMMARY

Senior engineer with 5+ years building production AI systems and the distributed data infrastructure that powers them, across telecommunications (Spectrum), financial services (Capital One), and cloud data platforms (Teradata). Currently building a multi-agent LangGraph diagnostic system that runs 8,000+ autonomous network investigations per day over a 50M+ endpoint network. Depth in Python and Go backend services, Kafka streaming, Kubernetes on AWS and Azure, and agent architecture: MCP tool integration, durable execution, evaluation, and observability. Track record of taking systems from design to production while measurably improving latency, reliability, and operational efficiency.

TECHNICAL SKILLS

AI / LLM Systems: LangGraph, LangChain, multi-agent (supervisor-worker) workflows, MCP tool integration, tool calling, agent state management, durable execution & checkpointing, RAG data preparation, agent tracing & evaluation, golden-dataset regression testing, guardrails (RBAC, schema validation, audit logging)

Languages: Python, Go, SQL, Bash

Data & Streaming: Apache Kafka, PostgreSQL, Redis, HBase, OpenSearch, OpenTSDB, ETL/ELT pipelines, stream & batch processing, data validation & quality monitoring

Distributed Systems: gRPC, REST, FastAPI, microservices, event-driven & asynchronous architectures, idempotency, deduplication, retry/recovery, state reconciliation, distributed caching, fault tolerance

Cloud & Infrastructure: AWS (EKS, S3), Azure (Blob Storage), Kubernetes, Docker, Helm, Terraform, CI/CD

Observability & Security: OpenTelemetry, distributed tracing, metrics & alerting, OAuth 2.0, mTLS, RBAC, PII protection, audit logging

PROFESSIONAL EXPERIENCE

Spectrum (Charter Communications) | Charlotte, NC

AI Engineer – Field Ops | SCOPE Platform | Mar 2025 – Present

- Designed and built a production multi-agent diagnostic system (Python, LangGraph) that autonomously troubleshoots faults across a 7-level HFC/DOCSIS 3.1 network hierarchy serving 50M+ subscriber endpoints; the system now handles 8,000+ investigations per day.
- Architected supervisor-worker agent workflows combining task decomposition, tool calling, state management, and recovery logic, reducing manual operational intervention in fault investigations by 35%.
- Integrated 20+ production tools and telemetry sources with the agents through MCP, enforcing RBAC, schema validation, authorization, and audit logging on every tool invocation so agent actions are controlled and traceable.
- Implemented durable execution and checkpointing on PostgreSQL and Redis, allowing long-running, multi-step investigations to resume from the last checkpoint after node or service failures instead of restarting.
- Built asynchronous telemetry processing handling 2M+ events per hour across HBase, OpenSearch, and OpenTSDB (15TB+ of network data at 99.9% completeness), giving agents real-time access to device health, signal metrics, and network state.
- Instrumented agent decisions, tool calls, workflow execution, and latency with OpenTelemetry distributed tracing; trace analysis across daily investigations contributed to an 18% improvement in root-cause accuracy.
- Developed fault-tolerant gRPC and asynchronous services with retries, timeouts, and failure recovery, sustaining 99.92% production availability during high-load scenarios.
- Migrated legacy monitoring services to containerized microservices on AWS EKS, reducing deployment cycles from ~2 weeks to under 1 day.

Technologies: Python, LangGraph, LangChain, MCP, Kafka, Kubernetes, AWS EKS, HBase, OpenSearch, OpenTSDB, Redis, PostgreSQL, gRPC, OpenTelemetry, Docker

Capital One | Dallas, TX

Data Engineer – Enterprise Data & Payment Intelligence | Jun 2023 – Mar 2025

- Built distributed data services for payment investigation workflows, processing ~3M financial and operational events per day through real-time and asynchronous pipelines and serving 25+ internal services and risk-analysis applications.
- Developed Apache Kafka pipelines processing ~1,200 events/sec with idempotent consumers, deduplication, retry handling, and recovery mechanisms, improving reliability of downstream processing.
- Reduced p95 query latency from 420ms to 170ms through Redis caching of hot account and transaction data (~82% hit rate) plus SQL query, index, and data-access tuning on high-volume transaction tables.
- Built Python/FastAPI services for ingesting, validating, transforming, and serving transaction data, with validation checks that catch duplicate events, malformed records, missing fields, and inconsistent transaction states before downstream processing.

- Prepared transaction, account, and operational datasets for retrieval in AI-powered financial investigation workflows, and integrated golden-dataset and trace-based regression tests into CI/CD to monitor data and AI workflow quality.
- Secured data-access services with OAuth 2.0, mTLS, RBAC, and PII protection, enabling secure data exchange across 25+ internal services.
- Added monitoring for Kafka throughput, consumer lag, processing failures, API latency, and data-quality issues, reducing production troubleshooting time by ~30%.
- Added monitoring for Kafka throughput, consumer lag, processing failures, API latency, and data-quality issues, reducing production troubleshooting time by approximately 30%.

Technologies: Python, SQL, Kafka, PostgreSQL, Redis, FastAPI, Kubernetes, AWS, REST APIs, OAuth 2.0, mTLS

Teradata | SanDiego, CA

Data Engineer – VantageCloud Data Platform | Aug 2021 – May 2023

- Developed high-concurrency Go microservices for the VantageCloud control plane, automating provisioning of 1,500+ multi-tenant database environments across AWS and Azure.
- Built reconciliation services in Go that compare desired vs. actual cloud resource state and self-correct drift, reducing provisioning failure rates by 12%.
- Optimized the Elastic Compute Provisioner for on-demand database workloads, reducing median provisioning latency from ~18 minutes to 11 minutes; Terraform and Kubernetes provisioning workflows cut compute deployment overhead by ~30%.
- Developed gRPC and REST APIs integrating Amazon S3 and Azure Blob Storage for backup, restore, and disaster-recovery workflows over ~40TB of backup data, increasing recovery throughput by ~25%.
- Implemented retry handling, state reconciliation, and resource-recovery logic to keep tenant environments consistent through cloud infrastructure failures and workload spikes.
- Raised automated provisioning and recovery coverage from ~60% to 85% through REST APIs and lifecycle automation, reducing manual provisioning work by ~40%.

Technologies: Go, SQL, AWS, Azure, Kubernetes, Terraform, Amazon S3, Azure Blob Storage, gRPC, REST APIs

EDUCATION

M.S. in Data Science and Analytics, New England University

CERTIFICATIONS

AWS Certified Solutions Architect – Professional
Microsoft Certified: Azure AI Engineer Associate